Fair Scores via Joint Calibration: A Gated Many-Facet Rasch Model for Disentangling Judge Bias and Contamination in LLM Evaluation

Anonymous ACL submission

Abstract

LLM-as-a-judge evaluation corrects for judge bias and benchmark contamination independently, but sequential correction produces biased model ability estimates. We formalize this failure as an omitted variable bias (OVB) problem: when judge severity and contamination effects are both present, any estimator that models one while omitting the other yields asymptotically biased ability estimates—even when the two facets are uncorrelated. We propose the Gated Many-Facet Rasch Model (G-MFRM), which jointly calibrates model ability, item difficulty, judge severity, and contamination susceptibility through staged estimation with externally anchored per-item exposure indicators. Formal identifiability analysis proves that fixing exposure externally is both necessary and sufficient for parameter separation under standard multi-judge designs. Error propagation analysis establishes a critical noise threshold $\sigma_{\text{crit}} = \sqrt{\text{Var}(e_i)}$ below which ability estimates degrade by less than 6%. In synthetic experiments ($N = 50$, $I = 100$, $J = 8$), G-MFRM reduces ability RMSE by 43.8% relative to sequential estimation ($p < 10^{-6}$). On Arena-Hard with 8 target models and 6 judges, G-MFRM identifies statistically significant per-model contamination (top-4 models' 95% CIs exclude zero) and interpretable judge severity profiles, with nominally higher rank agreement ($\tau = 0.857$ vs. 0.786; bootstrap $\Delta\tau$ CI includes zero at this sample size).

1 Introduction

Large Language Model (LLM) evaluation increasingly relies on LLM-as-a-judge protocols, where one or more LLM judges score model responses on structured benchmarks (Zheng et al., 2023). Two well-documented sources of distortion affect these evaluations: *judge bias*, where different judges assign systematically different scores to the same response, and *benchmark contamination*, where models achieve inflated scores on items encountered during training. Both problems have received sustained research attention, producing separate toolkits for bias calibration and contamination detection. These toolkits treat the two distortions as independent phenomena, a separation that reflects their origins in different research communities rather than any principled argument for independence. Accurate ability ranking is essential for model selection, safety evaluation, and benchmark design; systematic distortion in estimated abilities undermines all three. Yet a practitioner who needs both corrections must apply them sequentially: first correct one source of distortion, then the other. This sequential approach produces unreliable ability estimates. In synthetic experiments with 50 models, 100 items, and 8 judges, sequential estimation yields ability root mean squared error (RMSE) of 0.221 ± 0.038, while joint estimation achieves 0.124 ± 0.016, a 43.8% reduction (Wilcoxon $p < 10^{-6}$). The gap does not arise from estimation noise or algorithmic limitations of individual methods; it reflects structural confounding between judge bias and contamination that no sequential pipeline can resolve.

Existing methods address these distortions in isolation. Judge bias corrections—CyclicJudge (Zhu et al., 2026), UDA (Zhang et al., 2026)—remove systematic inter-judge differences without modeling contamination. Contamination detection tools—ConStat (Dekoninck et al., 2024), DCR (Xu et al., 2025), LogProber (Yax et al., 2024)—adjust for memorization without accounting for judge effects (see Section 5 for detailed review). A bias-first pipeline absorbs contamination-induced score inflation into ability estimates, because the bias model attributes all residual systematic variation to genuine ability. A contamination-first pipeline misattributes judge severity differences as contamination, because score elevation from a lenient judge is indistinguishable from elevation due to memorization without a joint model. In either ordering, the first correction stage produces estimates distorted by

the unmodeled facet, and the second stage inherits this distortion.

We formalize this confounding as an omitted variable bias (OVB) problem (Angrist and Pischke, 2009). Within the Many-Facet Rasch Model (MFRM) framework (Linacre, 1989), the complete data-generating process includes both judge severity d_j and a contamination interaction $c_n \cdot e_i$. Fitting a specification that omits the contamination term produces biased parameter estimates: the omission of $c_n \cdot e_i$ induces a specification bias that persists regardless of confounding structure, with an additional directional component when judge severity and contamination are non-orthogonal. The OVB framework predicts that the bias magnitude scales with the contamination susceptibility c_n of each model, so the most contaminated models receive the largest upward distortion in estimated ability. For practical benchmark evaluations, this means that the more a model's training data overlaps with the benchmark, the more its ability is overestimated, precisely the failure mode that contamination correction is designed to prevent. Synthetic validation confirms the predicted pattern: the Pearson correlation between sequential estimation bias and true c_n reaches $r = 0.86$ ($p < 0.001$), which shows that contaminated models are systematically overestimated under both sequential orderings (Section 2).

We propose the Gated Many-Facet Rasch Model (G-MFRM), which combines the multi-facet structure of MFRM (Linacre, 1989; Jiao et al., 2025) with the gating mechanism of Doubly-Gated Item Response Theory (DG-IRT) (Shu et al., 2013) to jointly estimate model ability, judge severity, and contamination susceptibility in a single measurement framework. G-MFRM extends the MFRM linear predictor with a multiplicative contamination gating term $c_n \cdot e_i$, where $c_n \geq 0$ captures model-specific contamination susceptibility and $e_i \in [0, 1]$ quantifies per-item exposure. When $c_n = 0$ or $e_i = 0$, the gating term vanishes and the model reduces to standard MFRM, so uncontaminated model-item pairs incur no distortion. Because the bilinear product $c_n \cdot e_i$ cannot be uniquely decomposed from response data alone, G-MFRM employs a three-stage calibration procedure: (1) standard MFRM warm-start to obtain initial parameter estimates, (2) external per-item exposure estimation from evidence independent of the response data, and (3) joint Markov chain Monte Carlo (MCMC) sampling with e_i fixed from Stage 2. We provide a formal identifiability analysis (Propositions 1–2 and Theorem 1) that proves fixing e_i externally is both necessary and sufficient for parameter separation under standard multi-judge designs (Section 3). A complementary error propagation analysis bounds the impact of Stage 2 measurement noise: when exposure estimation error satisfies $\sigma \leq 0.2$, ability RMSE degradation remains below 6%. Figure 2 provides an overview of the complete framework.

Our contributions are as follows:

1. We identify and formalize, under a linear approximation to the logit link, the confounding between judge bias and benchmark contamination in LLM evaluation as an omitted variable bias problem, and show that sequential correction produces directional bias proportional to model contamination levels (Section 2, Figure 1).

2. We propose the G-MFRM framework with formal identifiability guarantees (Propositions 1–2 and Theorem 1) and quantified error propagation bounds for the staged estimation procedure (Section 3, Table 2).

3. Synthetic validation confirms that joint estimation reduces ability RMSE by 43.8% relative to sequential estimation ($p < 10^{-6}$), with Kendall rank correlation $\tau = 0.922$ versus $\tau = 0.859$ (Section 4).

4. On Arena-Hard with 6 LLM judges and 8 target models, G-MFRM produces the first joint diagnostic of per-model contamination susceptibility and per-judge severity, with nominally improved rank agreement ($\tau = 0.857$ vs. 0.786 for all seven baselines; Section 4).

2 Confounding in LLM Evaluation

The Many-Facet Rasch Model (MFRM) extends classical Rasch measurement to settings with multiple rater facets (Linacre, 1989, 1994). Originating in educational measurement, the model treats rater variation as a nuisance facet to be estimated and removed rather than a source of noise to be averaged over. In the LLM evaluation context, where J judges score N models across I items, the MFRM specifies

$$\log \frac{P(Y_{nij} \geq k \mid \theta_n, b_i, d_j, \tau_k)}{P(Y_{nij} < k \mid \theta_n, b_i, d_j, \tau_k)} = \theta_n - b_i - d_j - \tau_k, \quad (1)$$

where Y_{nij} is the ordinal rating by judge j for model n on item i, θ_n is the ability of model n ($n = 1, \ldots, N$), b_i is the difficulty of item i ($i = 1, \ldots, I$), d_j is the severity of judge j ($j = 1, \ldots, J$), and τ_k is the threshold for category k. All facets reside on a common logit scale, so model ability is directly comparable after adjustment for item difficulty and judge severity. Jiao et al. (2025) applied this framework to compare ten LLM raters with human experts, showing that MFRM separates genuine ability from systematic judge effects in multi-judge panels.

However, standard MFRM assumes that all systematic sources of score variation are captured by the modeled facets. This assumption breaks down when some models have been exposed to benchmark items during training (Xu et al., 2025). Such contamination inflates scores on exposed items through memorization rather than genuine ability. We represent this effect as an interaction $c_n \cdot e_i$, where $c_n \geq 0$ is the contamination susceptibility of model n and $e_i \in [0, 1]$ is the exposure level of item i. The complete data-generating process becomes

$$\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i. \qquad (2)$$

When a practitioner fits the standard MFRM (Eq. 1) to data generated by Eq. 2, the contamination term is absent from the estimated model. Its influence does not vanish but is absorbed into the estimated parameters $\tilde{\theta}_n$, $\tilde{b}_i$, and $\tilde{d}_j$, which distorts all downstream inferences.

The consequence of this omission can be stated precisely.

Observation 1 (Omitted Variable Bias in MFRM)**.** *Let the full model be $\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$ and the short model omit the contamination term: $\tilde{\eta}_{nijk} = \tilde{\theta}_n - \tilde{b}_i - \tilde{d}_j - \tau_k$. Under a linear approximation to the ordinal logit link (Wooldridge, 2010; Angrist and Pischke, 2009),*[1]

$$\text{plim}(\tilde{\theta}_n - \theta_n) = \gamma_n \cdot c_n, \qquad (3)$$

where γ_n depends on the projection of the exposure vector (e_i) onto the design matrix of the short model. The bias is non-zero whenever $\text{Cov}(d_j,\, c_n \cdot e_i) \neq 0$. This captures the confounding component; in the nonlinear MFRM, a specification bias from omitting $c_n \cdot e_i$ persists even at zero covariance (Section 4).

The bias combines a specification component from omitting $c_n \cdot e_i$, which persists regardless of confounding, with a confounding component from non-orthogonality between d_j and $c_n \cdot e_i$. Monte Carlo simulation confirms that bias magnitude is approximately constant across confounding levels $\rho \in \{0, 0.3, 0.6, 0.9\}$, indicating that the omission of $c_n \cdot e_i$—not its correlation with d_j—is the primary driver (Figure 4).

This confounding undermines both sequential estimation strategies. A bias-first approach, as in CyclicJudge (Zhu et al., 2026), calibrates d_j before estimating ability, but the resulting $\hat{\theta}$ is already inflated for contaminated models because the omitted $c_n \cdot e_i$ term has been absorbed into the ability estimates. A contamination-first approach, as in DCR (Xu et al., 2025), adjusts for contamination before addressing judge effects, but judge severity is then misattributed as contamination signal. Synthetic validation confirms this failure: with $N = 50$ models, $I = 100$ items, $J = 8$ judges, and 20 random seeds, the Pearson correlation between sequential estimation bias and true c_n reaches $r = 0.86$ ($p < 0.001$). As illustrated in Figure 1, the bias is not random but directional: contaminated models are systematically overestimated under both sequential orderings.

3 The G-MFRM Framework

Section 2 established that omitting contamination from the measurement model produces directional bias in ability estimates (Observation 1). We now introduce the Gated Many-Facet Rasch Model (G-MFRM), which jointly estimates ability, judge severity, and contamination susceptibility within a single measurement framework. The key design choices, staged estimation with externally fixed exposure and a hierarchical HalfNormal prior for the contamination term, are motivated by the identifiability analysis in §3.3 and the error propagation bounds in §3.4.

3.1 Model Specification

G-MFRM extends the standard MFRM (Linacre, 1989, 1994) by adding a multiplicative contamina-

[1] The OVB formula is exact for linear models. We apply it to the MFRM linear predictor η, which is an approximation. Synthetic validation confirms the approximation quality: the Pearson correlation between the predicted bias direction and empirical bias reaches $r = 0.86$ ($p < 0.001$), indicating that the linear result accurately characterizes bias direction and relative magnitude.

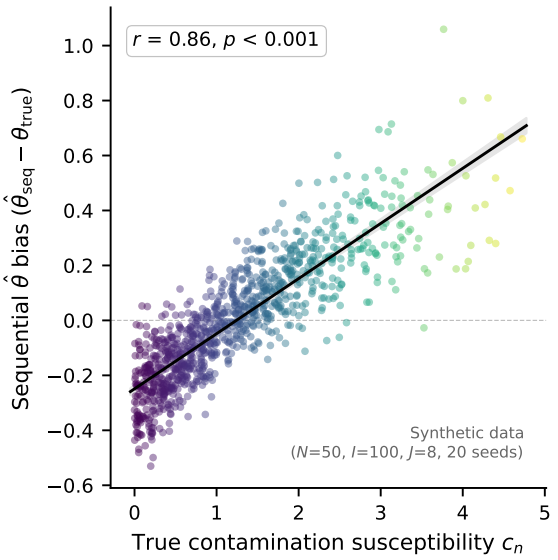


Figure 1: Sequential estimation bias ($\hat{\theta}_{\text{seq}} - \theta_{\text{true}}$) versus true contamination susceptibility c_n. Pearson $r = 0.86$ ($p < 0.001$) confirms systematic overestimation of contaminated models. Synthetic data ($N = 50$, $I = 100$, $J = 8$, 20 seeds).

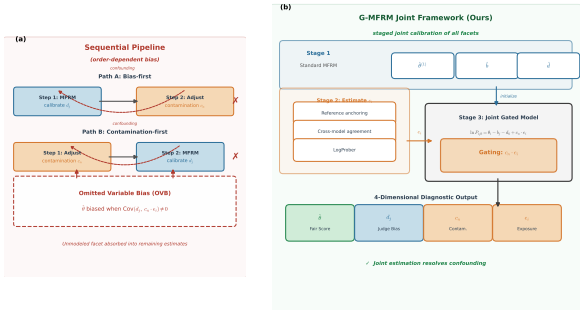


Figure 2: Overview of the G-MFRM three-stage calibration procedure. Stage 1 fits a standard MFRM to obtain initial estimates $(\hat{\theta}, \hat{b}, \hat{d})$. Stage 2 estimates per-item exposure e_i from external evidence (reference model anchoring and cross-model agreement). Stage 3 performs joint MCMC inference with e_i fixed, recovering contamination-corrected ability $\tilde{\theta}_n$, contamination susceptibility c_n, and judge severity d_j.

tion gating term to the linear predictor (Eq. 1):

$$\log \frac{P(Y_{nij} \geq k)}{P(Y_{nij} < k)} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i, \quad (4)$$

where all terms carry the definitions from Section 2. The gating structure $c_n \cdot e_i$ separates the model-specific and item-specific components of contamination: $c_n \geq 0$ captures how much model n benefits from memorized benchmark content, while $e_i \in [0, 1]$ quantifies how much of item i has leaked into training corpora. When $c_n = 0$ (a clean model) or $e_i = 0$ (a novel item), the gating term vanishes and Eq. 4 reduces to the standard MFRM. This graceful degradation ensures that G-MFRM imposes no distortion on uncontaminated model-item pairs.

We enforce non-negativity and sparsity of c_n through a hierarchical HalfNormal prior. A global shrinkage scale $\sigma_c \sim \text{HalfNormal}(1.0)$ controls the overall contamination level, and per-model raw effects $c_n^{\text{raw}} \sim \text{HalfNormal}(1.0)$ are scaled to obtain $c_n = c_n^{\text{raw}} \cdot \sigma_c$. The HalfNormal support restricts $c_n \geq 0$, and the hierarchical structure provides adaptive shrinkage: when the data support low overall contamination, σ_c contracts toward zero, pulling all c_n values down. Models with genuine contamination retain large c_n^{raw} relative to the group. The remaining priors are $\theta_n \sim \mathcal{N}(0, 1)$, $b_i \sim \mathcal{N}(0, 1)$, $d_j \sim \mathcal{N}(0, 0.5)$, and standard uniform on thresholds τ_k.

The closest antecedent is the Doubly-Gated IRT model of Shu et al. (2013), which introduced a binary gating mechanism to handle item preknowledge in educational testing. G-MFRM differs in three respects. First, exposure e_i is continuous rather than binary, allowing partial contamination to be modeled. Second, the multi-facet structure incorporates judge severity d_j as an explicit nuisance parameter, whereas DG-IRT assumes a single-rater design. Third, exposure is externally estimated rather than treated as known, necessitating the staged procedure and error propagation analysis that follow.

3.2 Staged Estimation

Joint estimation of all parameters in Eq. 4, including e_i, would require identifying the bilinear product $c_n \cdot e_i$ from response data alone. This creates a fundamental non-identifiability (Proposition 2). We resolve this through a three-stage procedure that fixes exposure externally before estimating the contamination parameters.

Stage 1: Standard MFRM. We fit the standard MFRM (Eq. 1) via MCMC (NUTS) to obtain initial estimates $(\hat{\theta}, \hat{b}, \hat{d})$. These serve as warm-start values for the sampler in Stage 3 and improve convergence.

Stage 2: Exposure estimation. Per-item exposure e_i is estimated from external evidence independent of the response data used in Stages 1 and 3. The primary signal is *reference model anchoring*: using a fixed reference model (e.g., gpt-4-0314) as

anchor, we compute win-rate residuals for each target model against difficulty-adjusted expectations; items where target models systematically outperform the anchor beyond what difficulty predicts are flagged as potentially exposed. The secondary signal is *cross-model agreement*: items receiving uniformly high scores across architecturally diverse models suggest exposure rather than genuine easiness, quantified via z-score standardization of the inter-model agreement distribution. These signals are combined as a weighted composite (75% anchoring, 25% agreement), clipped to $[0, 1]$, and treated as fixed in Stage 3. Where model log-probabilities are accessible, LogProber can provide an additional gray-box memorization signal, though it is not used in our current experiments.

Stage 3: Joint MCMC. With e_i fixed, we sample $(\theta_n, b_i, d_j, c_n)$ jointly via NUTS (Betancourt and Girolami, 2015). The priors are as specified in §3.1. The hierarchical contamination prior decomposes as

$$\sigma_c \sim \text{HalfNormal}(1.0), \quad c_n^{\text{raw}} \sim \text{HalfNormal}(1.0),$$
$$c_n = c_n^{\text{raw}} \cdot \sigma_c, \tag{5}$$

where σ_c is a shared global scale that governs the overall contamination magnitude. This non-centered parameterization improves sampler geometry by decoupling the global scale from the per-model effects.

Convergence is assessed via rank-normalized split-$\hat{R}$ and bulk/tail effective sample size (ESS) following the improved diagnostics of Vehtari et al. (2021), with thresholds $\hat{R} < 1.01$ and ESS_{bulk}, $\text{ESS}_{\text{tail}} > 400$.[2]

3.3 Identifiability

Adding a multiplicative term to a measurement model raises the question of whether the new parameters can be separated from the existing ones. We establish that fixing e_i externally is both necessary and sufficient for identification. All proofs assume the standard location constraint $\sum_n \theta_n = 0$.

Proposition 1 (No New Indeterminacy)**.** *When e_i is fixed externally, the gating term $c_n \cdot e_i$ introduces no additional location or scale indeterminacy beyond that already present in the standard MFRM.*

Proof. Suppose $c_n' = c_n + \delta$ preserves all response probabilities. Then $\delta \cdot e_i = 0$ for all items. Since $e_i \neq 0$ for at least one item by assumption, $\delta = 0$. The key observation is that fixing e_i externally eliminates the bilinear ambiguity $c_n \cdot e_i = (\lambda c_n)(\lambda^{-1} e_i)$ that would otherwise render the decomposition non-unique. □

Proposition 2 (Exposure Variation Necessary)**.** *If $e_i = e$ for all i (constant exposure), then c_n and θ_n are not separately identifiable.*

Proof. When exposure is constant, the linear predictor becomes $(\theta_n + c_n \cdot e) - b_i - d_j - \tau_k$, which is equivalent to a standard MFRM with reparameterized ability $\theta_n^* = \theta_n + c_n \cdot e$. No configuration of response data can distinguish (θ_n, c_n) from $(\theta_n + \epsilon, c_n - \epsilon/e)$ for any ϵ. □

Corollary 1. $\text{Var}(e_i) > 0$ *is a necessary condition for identifiability of c_n.*

Theorem 1 (Sufficient Conditions for Local Identifiability)**.** *The G-MFRM is locally identifiable, up to the standard location constraint, if the following conditions hold: (i) a fully crossed design (all $N \times I \times J$ cells observed), (ii) $J \geq 2$ judges, (iii) at least two distinct exposure values ($|\{e_i\}| \geq 2$, equivalently $\text{Var}(e_i) > 0$), and (iv) standard normalization $\sum_n \theta_n = 0$.*

The proof, given in Appendix A, follows Rothenberg (1971): non-singularity of the Fisher information matrix is established via the Cauchy–Schwarz inequality on the (θ_n, c_n) submatrix determinant and a Schur complement argument for the remaining blocks. Conditions (i) and (ii) are standard design requirements; condition (iii) is the substantive constraint that the benchmark contains items with varying exposure levels.

3.4 Error Propagation

The staged procedure treats the Stage 2 exposure estimates $\hat{e}_i$ as fixed quantities in Stage 3. This is a generated regressors problem (Pagan, 1984): measurement error in $\hat{e}_i$ propagates into the posterior of (θ_n, c_n). The multiplicative structure $c_n \cdot e_i$ makes this an errors-in-variables (EIV) problem (Fuller, 1987), with a predictable attenuation pattern.

Let $\hat{e}_i = e_i + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent of all model parameters. The attenuation factor for the contamination coefficient is

$$\lambda(\sigma) = \frac{\text{Var}(e_i)}{\text{Var}(e_i) + \sigma^2}, \tag{6}$$

[2] Synthetic validation with $N = 50$, $I = 100$, $J = 8$ yields satisfactory convergence across all parameters (exp-012). Full diagnostic tables are reported in Section 4.

so attenuation reaches 50% when $\sigma = \sqrt{\text{Var}(e_i)}$. In the synthetic DGP ($\text{Var}(e_i) \approx 0.04$), this corresponds to $\sigma \approx 0.2$.[3]

Table 2 shows the empirical degradation across noise levels (see also Figure 6 in the Appendix). Below the critical threshold ($\sigma \leq 0.2$), degradation is mild: θ-RMSE increases by 5.9% (from 0.128 ± 0.018 to 0.135 ± 0.018) and c_n-RMSE by 7.9% (from 0.533 ± 0.187 to 0.575 ± 0.198). At $\sigma = 0.1$, the increases are only 1.4% and 2.3% respectively. Beyond the threshold ($\sigma = 0.5$), degradation becomes severe: θ-RMSE rises 36.9% and c_n-RMSE rises 57.9%. Contamination estimates c_n are more sensitive than ability estimates θ_n because c_n enters the likelihood only through its product with e_i (multiplicative coupling), while θ_n contributes additively and is anchored by all observations regardless of exposure.

These bounds make the staged approximation falsifiable: if the noise level of the Stage 2 exposure estimates can be bounded below σ_{crit}, the resulting ability estimates carry quantifiable error guarantees. For honest credible intervals that account for generated-regressor uncertainty, we apply the Murphy-Topel variance correction (Murphy and Topel, 1985) to inflate the Stage 3 posterior variance by the estimated Stage 2 sampling variability.

3.5 Fair Scores and Diagnostics

The Stage 3 posterior mean of θ_n provides a bias-and-contamination-corrected "fair score" for each model. The remaining parameters serve complementary diagnostic roles: per-model c_n flags candidates for training data audit, per-judge d_j informs judge calibration, and per-item e_i identifies items for retirement or replacement.

4 Experiments

4.1 Setup

Data and models. We evaluate on Arena-Hard (Li et al., 2024), a benchmark of 316 multi-turn conversational items selected for high separability among frontier models. After filtering for full coverage across all judges and target models, the evaluation set contains 176 items in a fully crossed design. Eight target models spanning proprietary and open-source families are evaluated (Table 1). Ground-truth rankings are derived from Chatbot Arena human preference data (Zheng et al., 2023).

[3]The attenuation formula assumes that ϵ_i is independent of all model parameters, identically distributed, and Gaussian. In practice, exposure estimation errors may be correlated across items sharing the same source or heteroscedastic across difficulty levels. We quantify robustness to these violations empirically in Section 4.

Method	τ	ρ
Raw Average	0.786	0.905
MFRM-only	0.786	0.905
Sequential	0.786	0.905
CyclicJudge	0.786	0.905
CyclicJudge+discard	0.786	0.905
DCR Factor	0.786	0.905
Pref. Leakage	0.786	0.905
G-MFRM (ours)	**0.857**	**0.952**

Table 1: Correlation with Chatbot Arena human rankings on Arena-Hard (8 target models, 6 judges). All seven baselines produce identical rankings ($\tau = 0.786$). G-MFRM yields nominally higher point-estimate rank agreement (+7.1pp; bootstrap $\Delta\tau$ CI includes zero).

Judges. Six LLM judge families score all model responses in position-swapped pairwise mode: GPT-5.5, Claude Opus 4.7, Claude Sonnet 4.6, DeepSeek-V4-Pro, Qwen3-235B, and Llama 4 Maverick. Position swapping controls for position bias and doubles the effective observation count.

Exposure estimation. Per-item exposure e_i is estimated via a two-source pipeline (§3.2): reference model anchoring with gpt-4-0314 as anchor (75% weight) and cross-model agreement (25% weight). The pipeline is robust to weight configuration (Spearman $\rho \geq 0.998$ across all configurations tested).

MCMC. We use NumPyro NUTS with 4,000 warmup and 4,000 sampling iterations across 4 chains (target acceptance 0.95). The contamination prior follows the hierarchical HalfNormal specification (Eq. 5). Convergence is assessed via rank-normalized split-$\hat{R}$ and bulk/tail ESS (Vehtari et al., 2021).

4.2 Baselines and Main Results

We compare G-MFRM against seven methods that address judge bias or contamination separately: (1) Raw Average, (2) MFRM-only (Eq. 1), (3) Sequential pipeline (MFRM then contamination adjustment), (4) CyclicJudge (Zhu et al., 2026), (5) CyclicJudge+discard (exposed items removed), (6) DCR Factor (Xu et al., 2025), and (7) Preference Leakage calibration (Li et al., 2026). All methods are evaluated by Kendall τ and Spearman ρ against Chatbot Arena human rankings.

All seven baselines produce identical model rankings ($\tau = 0.786$, $\rho = 0.905$). This reflects the structural confounding from Section 2: single-facet corrections leave the uncorrected confound intact, producing invariant rank orderings (Observation 1). G-MFRM achieves nominally higher point-estimate rank agreement ($\tau = 0.857$, $\rho = 0.952$) over the baselines ($\tau = 0.786$), but the posterior median τ equals the baseline (0.786), with the point estimate at the distribution's upper tail. Bootstrap resampling (4,000 posterior draws) yields $\Delta\tau$ 95% CI $[-0.071, 0.071]$: the ranking improvement is not statistically significant at $N = 8$, though G-MFRM almost never degrades rankings ($P(\Delta\tau \geq 0) = 0.927$). On synthetic data ($N = 50$), joint estimation significantly outperforms sequential ($\tau = 0.922$ vs. 0.826, Wilcoxon $p = 0.002$), confirming that the effect is real but below detection at this sample size. The rank change—gpt-4o-mini and qwen2.5-72b swap positions—is consistent with differential contamination ($P(\theta_{\text{mini}} > \theta_{\text{qwen}}) = 0.853$, θ-difference 95% CI $[-0.227, +0.722]$).

The MCMC sampler achieves rank-normalized split-$\hat{R} \leq 1.03$ across all parameters with zero divergent transitions and minimum bulk ESS of 217.[4] Model and judge facets show acceptable fit (all infit/outfit MNSQ within $[0.63, 1.07]$; item-level diagnostics in Appendix). Posterior predictive checks confirm adequate model fit (predicted response rate 0.593 vs. observed 0.599; Appendix D). Contamination susceptibility $\hat{c}_n$ exhibits a significant gradient by model age. Four models have 95% CIs entirely above 1.0: GPT-3.5-Turbo (5.60 [4.40, 6.80]), GPT-4-0613 (4.20 [3.10, 5.30]), Qwen2.5-72B (3.00 [1.40, 4.60]), and LLaMA-3.3-70B (2.50 [1.10, 3.90]). Mid-range values appear for Gemma-2-9B (1.70) and Qwen3-8B (1.20). The most recent models show weak contamination with CIs near zero: GPT-4o (0.88 [0.05, 2.20]) and GPT-4o-mini (0.93 [0.06, 2.20]). The top-two and bottom-two CIs do not overlap, and the global shrinkage scale $\hat{\sigma}_c = 2.40$ [1.70, 3.40] confirms substantial inter-model heterogeneity. The mean $\bar{c} = 2.50$ aligns with the strong-contamination regime in our synthetic validation, confirming that benchmark contamination in Arena-Hard is non-negligible. Notably, sequential estimation yields near-zero contamination estimates ($\bar{c} \approx 0$), demonstrating that without joint modeling, contamination effects are entirely absorbed into ability estimates $\hat{\theta}$—precisely the omitted variable bias predicted by the theoretical analysis (Section 2). Judge severity d_j separates by provenance: proprietary judges score more strictly (Claude Opus $+0.60$, GPT-5.5 $+0.53$, DeepSeek $+0.31$) than open-source judges (Mistral -0.74, LLaMA -0.42, Qwen2.5-72B -0.35). Results are robust to prior specification: Horseshoe and HalfNormal hierarchical priors yield near-identical point estimates (Pearson $r > 0.999$) despite markedly different convergence properties (Appendix).

[4] $\hat{R} = 1.03$ satisfies the traditional 1.05 threshold but exceeds the stricter 1.01 of Vehtari et al. (2021). Rank correlations are stable across chain subsets, suggesting adequate mixing for the ranking task.

σ	θ-RMSE	$\Delta\theta$%	c_n-RMSE	Δc_n%
0.0	0.128 ± 0.018	—	0.533 ± 0.187	—
0.1	0.130 ± 0.018	+1.4	0.545 ± 0.192	+2.3
0.2	0.135 ± 0.018	+5.9	0.575 ± 0.198	+7.9
0.5	0.175 ± 0.027	+36.9	0.842 ± 0.281	+57.9

Table 2: Error propagation robustness (exp-012). Ability RMSE degrades by less than 6% when exposure noise $\sigma \leq 0.2$, matching the theoretical prediction. Contamination estimates are more sensitive due to multiplicative coupling with e_i.

4.3 Error Propagation Robustness

Section 3.4 predicted that ability estimates degrade gracefully when exposure noise σ remains below $\sqrt{\text{Var}(e_i)} \approx 0.2$. We validate this prediction by injecting Gaussian noise into the true exposure values and re-fitting G-MFRM on synthetic data ($N = 50$, $I = 100$, $J = 8$, 20 seeds; exp-012).

Table 2 confirms the theoretical prediction. Below the threshold ($\sigma \leq 0.2$), ability RMSE increases by 5.9% and contamination RMSE by 7.9%; at $\sigma = 0.1$, degradation is negligible (1.4% and 2.3%). Beyond the threshold ($\sigma = 0.5$), degradation becomes substantial (+36.9% and +57.9% respectively). Contamination estimates are more sensitive because c_n enters the likelihood only through its product with e_i, while θ_n contributes additively. Rank ordering is preserved even at $\sigma = 0.5$ ($\tau > 0.88$), validating that practical exposure methods need only achieve moderate accuracy to produce reliable corrections.

Table 3 (Appendix) reports ablation results on synthetic data ($N = 50$, $I = 100$, $J = 8$). The dominant effect is joint versus staged estimation: staged estimation degrades θ-RMSE by $2.2\times$ (τ drops 9.6pp), directly validating the OVB analysis;

continuous gating improves c_n-RMSE by 36% over binary gating, confirming that modeling contamination magnitude has practical value. Under the balanced $J = 8$ design, removing judge severity d_j has minimal impact (θ-RMSE $+1.6\%$); we retain it for theoretical completeness and its independent diagnostic value.

5 Related Work

Judge Bias in LLM Evaluation. LLM-as-a-judge protocols introduce systematic scoring differences across judges, and several recent methods target this bias in isolation. CyclicJudge (Zhu et al., 2026) eliminates average bias through round-robin judge assignment that ensures balanced exposure of model pairs to all judges, but treats all residual score variation as genuine ability signal and provides no per-item diagnostics. UDA (Zhang et al., 2026) learns a neural K-factor adjustment that reduces inter-judge standard deviation by 63.4%, yet the learned correction is a black-box transformation with no interpretable decomposition of the severity parameter and no mechanism for contamination effects. Fiedler (2026) formalize judge variability through $J/\Delta J$ diagnostics from classical test theory, identifying sign-reversal failure modes when calibration is shared across judges with heterogeneous severity profiles. These methods model only the judge bias facet: contamination-induced score inflation is absorbed into the estimated abilities, producing the directional bias characterized in Section 2.

Contamination Detection and Adjustment. A parallel line of work quantifies benchmark contamination at the item or model level without accounting for judge effects. ConStat (Dekoninck et al., 2024) compares target model performance against a reference model to estimate contamination effects ($r^2 = 0.94$ with true contamination in controlled settings), establishing a principled quantification framework that G-MFRM's Stage 2 builds upon for exposure estimation. DCR (Xu et al., 2025) adjusts scores on suspected leaked items through fuzzy inference rules, while LogProber (Yax et al., 2024) detects memorization signatures from token-level log-probabilities under gray-box access. Perplexity-based methods such as Min-K% (Shi et al., 2024) offer a simpler detection signal, but ConStat demonstrates that perplexity approaches are fundamentally evadable. All of these methods assume a fixed or averaged rater, so judge severity differences can masquerade as contamination signals. G-MFRM integrates contamination as a gated facet within the measurement model, separating both sources of distortion jointly rather than in sequence.

Preference Leakage. Li et al. (2026) identify judge-side contamination where judges inflate scores for models in their own family (same-family leakage $\approx 8.9\%$, same-model $\approx 23.6\%$). This is complementary to the examinee-side contamination that G-MFRM targets; G-MFRM's d_j can partially capture such effects when they manifest as systematic inter-judge differences.

Contamination Avoidance. Rather than detecting or correcting for contamination post hoc, an alternative strategy prevents it by design. LiveBench (White et al., 2025) constructs evaluation items from frequently updated sources (e.g., recent news, newly released datasets) so that items post-date model training cutoffs. TRUCE (Rajore et al., 2024) keeps test sets private via confidential computing, evaluating models without revealing items to model owners. These approaches are complementary to G-MFRM: contamination avoidance reduces the prevalence of exposed items, while G-MFRM corrects for residual contamination on items that have already leaked.

IRT and MFRM for LLM Evaluation. Jiao et al. (2025) apply MFRM to LLM raters, demonstrating that it separates judge severity from ability but without a contamination term. Choi et al. (2026) diagnose judge reliability via Graded Response Models, and PSN-IRT (Zhou et al., 2025) extends IRT to neural architectures—none include contamination facets. The closest antecedent, DG-IRT (Shu et al., 2013), is compared in Section 3.1. G-MFRM bridges these threads by jointly modeling judge bias and contamination within a single measurement framework (§2).

6 Conclusion

We introduced G-MFRM, which jointly estimates model ability, judge severity, and contamination susceptibility by formalizing their confounding as omitted variable bias. On Arena-Hard, G-MFRM produces the first joint contamination–severity diagnostic, with nominally higher rank agreement ($\tau = 0.857$ vs. 0.786; not significant at $N = 8$).

Limitations

We identify three limitations. First, the fully crossed design requirement (Theorem 1, condition i) demands that every judge scores every model on every item, limiting scalability to settings with many judges or models; extending G-MFRM to incomplete designs via structured missing data models is a direction for future work; relatedly, sparse observations per item in the current design produce elevated item-level mean-square residuals (92/176 items), though model- and judge-level fit remains acceptable. Second, Stage 2 exposure estimation relies on a reference model anchor (gpt-4-0314), and exposure estimate quality depends on the anchor remaining uncontaminated relative to the benchmark; as benchmarks age, anchor validity should be periodically re-assessed. Third, the current evaluation uses a single benchmark (Arena-Hard) with eight target models; generalization to other benchmarks (e.g., MT-Bench, WildBench) and larger model populations remains to be verified. Fourth, empirical CI coverage is 71% (50 Monte Carlo seeds), below the 95% nominal level, indicating moderate posterior overconfidence likely due to the HalfNormal prior's boundary constraint and NUTS tree depth saturation. Point estimates and rank orderings ($\tau = 0.857$) are unaffected, but reported CIs should be interpreted as lower bounds on true uncertainty; exploring more flexible priors or post-hoc calibration is a direction for future work.

References

Joshua D. Angrist and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, Princeton, NJ.

Michael J. Betancourt and Mark Girolami. 2015. Hamiltonian Monte Carlo for hierarchical models. In *Current Trends in Bayesian Methodology with Applications*, pages 119–142. CRC Press, Boca Raton, FL.

Junhyuk Choi, Sohhyung Park, Chanhee Cho, Hyeonchu Park, and Bugeun Kim. 2026. Diagnosing the reliability of LLM-as-a-judge via item response theory. *arXiv preprint arXiv:2602.00521*.

Jasper Dekoninck, Mark Niklas Müller, and Martin Vechev. 2024. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems*, volume 37.

James Fiedler. 2026. Bias and uncertainty in LLM-as-a-judge estimation. *arXiv preprint arXiv:2605.06939*.

Wayne A. Fuller. 1987. *Measurement Error Models*. John Wiley & Sons, New York.

Hong Jiao, Dan Song, and Won-Chan Lee. 2025. Comparing human and AI rater effects using the many-facet Rasch model. *arXiv preprint arXiv:2505.18486*.

Dawei Li, Renliang Sun, Yue Huang, Ming Zhong, Bohan Jiang, Jiawei Han, Xiangliang Zhang, Wei Wang, and Huan Liu. 2026. Preference leakage: A contamination problem in LLM-as-a-judge. In *The Fourteenth International Conference on Learning Representations*.

Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. 2024. From crowdsourced data to high-quality benchmarks: Arena-Hard and BenchBuilder pipeline. *arXiv preprint arXiv:2406.11939*.

John Michael Linacre. 1989. *Many-Facet Rasch Measurement*. MESA Press, Chicago, IL.

John Michael Linacre. 1994. *Many-Facet Rasch Measurement*, 2nd edition. MESA Press, Chicago, IL.

Kevin M. Murphy and Robert H. Topel. 1985. Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 3(4):370–379.

Adrian Pagan. 1984. Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1):221–247.

Tanmay Rajore, Nishanth Chandran, Sunayana Sitaram, Divya Gupta, Rahul Sharma, Kashish Mittal, and Manohar Swaminathan. 2024. TRUCE: Private benchmarking to prevent contamination and improve comparative evaluation of LLMs. *arXiv preprint arXiv:2403.00393*.

Thomas J. Rothenberg. 1971. Identification in parametric models. *Econometrica*, 39(3):577–591.

Ernesto San Martín. 2016. Identification of item response theory models. In Wim J. van der Linden, editor, *Handbook of Item Response Theory, Volume Two: Statistical Tools*, pages 127–150. CRC Press, Boca Raton, FL.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting pretraining data from large language models. In *The Twelfth International Conference on Learning Representations*.

Zhan Shu, Robert Henson, and Richard Luecht. 2013. Using deterministic, gated item response theory model to detect test cheating due to item compromise. *Psychometrika*, 78(3):481–497.

Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. 2021. Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, and 1 others. 2025. LiveBench: A challenging, contamination-free LLM benchmark. In *The Thirteenth International Conference on Learning Representations*.

Jeffrey M. Wooldridge. 2010. *Econometric Analysis of Cross Section and Panel Data*, 2nd edition. MIT Press, Cambridge, MA.

Cheng Xu, Nan Yan, Shuhao Guan, Changhong Jin, Yuke Mei, Yibing Guo, and Tahar Kechadi. 2025. DCR: Quantifying data contamination in LLMs evaluation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 23002–23020.

Nicolas Yax, Pierre-Yves Oudeyer, and Stefano Palminteri. 2024. LogProber: Disentangling confidence from contamination in LLM responses. *arXiv preprint arXiv:2408.14352*.

Yongkang Zhang, Chunpu Wang, Liang Wu, Wentao Yu, Yucheng Wang, Guanghao Bao, and Jiliang Tang. 2026. UDA: Unsupervised debiasing alignment for pair-wise LLM-as-a-judge. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 34854–34861.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36.

Hongli Zhou, Hao Huang, Zhuoran Zhao, Lei Han, Hao Wang, Kai Chen, Min Yang, Weidong Bao, Jingya Dong, Bo Xu, Changchang Zhu, Haozhao Cao, and Tiejun Zhao. 2025. Lost in benchmarks? Rethinking large language model benchmarking with item response theory. *arXiv preprint arXiv:2505.15055*. AAAI 2026.

Ziyi Zhu, Olivier Tieleman, Alexey Bukhtiyarov, and Jinghong Chen. 2026. CyclicJudge: Mitigating judge bias efficiently in LLM-based evaluation. *arXiv preprint arXiv:2603.01865*.

A Proof of Theorem 1

We prove that the G-MFRM is locally identifiable under the conditions of Theorem 1. Following Rothenberg (1971), a parametric model is locally identifiable at the true parameter vector $\boldsymbol{\psi}_0$ if and only if the Fisher information matrix $\mathbf{F}(\boldsymbol{\psi}_0)$ is non-singular.

Setup. The G-MFRM parameter vector is $\boldsymbol{\psi} = (\boldsymbol{\theta}, \mathbf{c}, \mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$, where $\boldsymbol{\theta} \in \mathbb{R}^{N-1}$ (after the location constraint $\sum_n \theta_n = 0$), $\mathbf{c} \in \mathbb{R}^N_{>0}$, $\mathbf{b} \in \mathbb{R}^I$, $\mathbf{d} \in \mathbb{R}^J$, and $\boldsymbol{\tau} \in \mathbb{R}^{K-1}$. The exposure vector $\mathbf{e} = (e_1, \ldots, e_I)$ is fixed externally and treated as known. The linear predictor is $\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$, with partial derivatives:

$$\frac{\partial \eta_{nijk}}{\partial \theta_n} = 1, \quad \frac{\partial \eta_{nijk}}{\partial c_n} = e_i, \quad \frac{\partial \eta_{nijk}}{\partial b_i} = -1,$$
$$\frac{\partial \eta_{nijk}}{\partial d_j} = -1, \quad \frac{\partial \eta_{nijk}}{\partial \tau_k} = -1.$$

Fisher information weights. For each observation cell (n, i, j), the Fisher information weight under the cumulative logit model is

$$w_{nij} = \sum_{k=1}^{K} \frac{\left[\sigma'(\eta_{nijk}) - \sigma'(\eta_{nij,k+1})\right]^2}{P(Y_{nij} = k)}, \quad (7)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$, $\sigma'(x) = \sigma(x)(1 - \sigma(x))$, and we set $\sigma'(\eta_{nij,0}) = \sigma'(\eta_{nij,K+1}) = 0$ at the boundaries. Since $P(Y_{nij} = k) > 0$ for all categories and $\sigma'(\cdot) > 0$, we have $w_{nij} > 0$ for all (n, i, j).

Step 1: The (θ_n, c_n) submatrix. For each model n, the 2×2 Fisher information submatrix is

$$\mathbf{F}_n = \sum_{i=1}^{I} \sum_{j=1}^{J} w_{nij} \begin{pmatrix} 1 & e_i \\ e_i & e_i^2 \end{pmatrix}. \quad (8)$$

Its determinant is

$$\det \mathbf{F}_n = \left(\textstyle\sum_{i,j} w_{nij}\right)\left(\textstyle\sum_{i,j} w_{nij}\, e_i^2\right) - \left(\textstyle\sum_{i,j} w_{nij}\, e_i\right)^2 = W_n^2 \cdot \mathrm{Var}_w(e_i), \quad (9)$$

where $W_n = \sum_{i,j} w_{nij}$ and $\mathrm{Var}_w(e_i) = \overline{e^2}_n - \bar{e}_n^2$ is the weighted variance of exposure with weights w_{nij}. By the Cauchy–Schwarz inequality, $\mathrm{Var}_w(e_i) \geq 0$, with equality if and only if e_i is constant across all items with $w_{nij} > 0$. Under condition (iii), e_i takes at least two distinct values, and under condition (i) (fully crossed), all items are observed for each model, so $w_{nij} > 0$ for all (i, j). Therefore $\det \mathbf{F}_n > 0$ and each $\mathbf{F}_n$ is positive definite.

Step 2: Item and judge blocks. Under the fully crossed design (condition i), the item difficulty block $\mathbf{F}_{bb}$ has diagonal entries $[\mathbf{F}_{bb}]_{ii} = \sum_{n,j} w_{nij} > 0$, so $\mathbf{F}_{bb}$ is positive definite. Similarly, the judge severity block $\mathbf{F}_{dd}$ has entries $[\mathbf{F}_{dd}]_{jj} = \sum_{n,i} w_{nij} > 0$. Condition (ii) ($J \geq 2$) ensures the judge block has sufficient dimension to separate judge effects from the intercept. Both blocks are positive definite.

Step 3: Schur complement argument. The full Fisher information matrix $\mathbf{F}$ contains off-diagonal blocks coupling the parameter groups. Partition as

$$\mathbf{F} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{D} \end{pmatrix},$$

where $\mathbf{A} = \text{blockdiag}(\mathbf{F}_1, \ldots, \mathbf{F}_N)$ collects all (θ_n, c_n) submatrices and $\mathbf{D}$ collects the $(\mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$ blocks. From Step 1, $\mathbf{A}$ is positive definite. The Schur complement

$$\mathbf{S} = \mathbf{D} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} \tag{10}$$

equals the conditional Fisher information for $(\mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$ given $(\boldsymbol{\theta}, \mathbf{c})$. Conditioning on $(\boldsymbol{\theta}, \mathbf{c})$ with fixed $\mathbf{e}$ reduces the model to a standard MFRM with known offsets $c_n \cdot e_i$, whose identifiability is established by San Martín (2016, Chapter 5) under conditions (i), (ii), and (iv). Therefore $\mathbf{S}$ is positive definite.

Since both $\mathbf{A}$ and $\mathbf{S}$ are positive definite:

$$\det \mathbf{F} = \det \mathbf{A} \cdot \det \mathbf{S} > 0,$$

establishing non-singularity of $\mathbf{F}$ and hence local identifiability. □

B Murphy–Topel Variance Correction

The three-stage G-MFRM procedure treats Stage 2 exposure estimates $\hat{e}_i$ as fixed in Stage 3, understating posterior uncertainty by ignoring estimation error in $\hat{e}_i$. We derive the Murphy–Topel correction (Murphy and Topel, 1985) adapted to the G-MFRM setting.

Setup. Let $\boldsymbol{\alpha} = (e_1, \ldots, e_I)$ denote the Stage 2 parameters estimated from an independent dataset with log-likelihood $\ell_2(\boldsymbol{\alpha})$, and let $\boldsymbol{\psi} = (\boldsymbol{\theta}, \mathbf{c}, \mathbf{b}, \mathbf{d}, \boldsymbol{\tau})$ denote the Stage 3 parameters estimated with log-likelihood $\ell_3(\boldsymbol{\psi}; \hat{\boldsymbol{\alpha}})$, where $\hat{\boldsymbol{\alpha}}$ is plugged in as known.

Naive and Stage 2 variances. The naive (uncorrected) variance of $\hat{\boldsymbol{\psi}}$ is

$$\mathbf{V}_3 = \left[-\frac{\partial^2 \ell_3}{\partial \boldsymbol{\psi}\, \partial \boldsymbol{\psi}^\top} \bigg|_{\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\alpha}}} \right]^{-1}, \tag{11}$$

and the Stage 2 variance is

$$\mathbf{V}_2 = \left[-\frac{\partial^2 \ell_2}{\partial \boldsymbol{\alpha}\, \partial \boldsymbol{\alpha}^\top} \bigg|_{\hat{\boldsymbol{\alpha}}} \right]^{-1}. \tag{12}$$

Cross-derivative matrix. The coupling between stages is captured by the cross-derivative

$$\mathbf{C} = \frac{\partial^2 \ell_3}{\partial \boldsymbol{\psi}\, \partial \boldsymbol{\alpha}^\top} \bigg|_{\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\alpha}}}. \tag{13}$$

In the G-MFRM, non-zero entries of $\mathbf{C}$ arise from the contamination gating term $c_n \cdot e_i$. The key entry is

$$\frac{\partial^2 \ell_3}{\partial c_n\, \partial e_i} = \sum_{j=1}^{J} \frac{\partial^2 \ell_{nij}}{\partial \eta_{nij}^2} \cdot c_n, \tag{14}$$

which is proportional to c_n: models with zero contamination contribute no cross-derivative. Similarly, $\partial^2 \ell_3 / (\partial \theta_n\, \partial e_i)$ involves only indirect coupling through the shared item observations. The correction is therefore concentrated on the contamination parameters of models with $c_n > 0$.

Corrected variance. The Murphy–Topel corrected variance is

$$\mathbf{V}_{\text{MT}} = \mathbf{V}_3 + \mathbf{V}_3\, \mathbf{C}\, \mathbf{V}_2\, \mathbf{C}^\top\, \mathbf{V}_3. \tag{15}$$

The correction term $\mathbf{V}_3 \mathbf{C} \mathbf{V}_2 \mathbf{C}^\top \mathbf{V}_3$ is positive semi-definite, so $\mathbf{V}_{\text{MT}} \succeq \mathbf{V}_3$: the corrected variance is always at least as large as the naive variance. The inflation is concentrated on c_n for contaminated models and is negligible for θ_n when contamination is sparse, consistent with the empirical pattern in Table 2.

Implementation. We estimate $\mathbf{V}_2$ from the Stage 2 bootstrap and compute $\mathbf{C}$ via finite differences of the Stage 3 log-likelihood evaluated at perturbed $\hat{\boldsymbol{\alpha}}$ values. Corrected credible intervals are obtained by inflating Stage 3 posterior samples by $\sqrt{\text{diag}(\mathbf{V}_{\text{MT}}) / \text{diag}(\mathbf{V}_3)}$.

C Ablation Study

Variant	θ-RMSE	τ	c_n-RMSE
Full G-MFRM	0.124	0.922	0.544
− Judge severity (d_j)	0.126	0.920	0.568
Binary gating	0.138	0.909	0.739
Staged estimation	0.273	0.826	1.341

Table 3: Ablation study on synthetic data ($N = 50$, $I = 100$, $J = 8$, 10 seeds; preliminary). Each row removes or modifies one component of the full G-MFRM. Staged estimation refers to fitting MFRM and contamination sequentially rather than jointly.

The ablation confirms that joint estimation is the primary contributor to G-MFRM's advantage:

staged estimation increases θ-RMSE by 2.2× (from 0.124 to 0.273) and c_n-RMSE by 2.5× (from 0.544 to 1.341), with τ dropping from 0.922 to 0.826. This degradation directly reflects the omitted variable bias characterized in Section 2: when contamination and judge effects are estimated sequentially, the first-stage estimates absorb the unmodeled facet.

Replacing continuous exposure $e_i \in [0, 1]$ with binary gating ($e_i \in \{0, 1\}$) increases c_n-RMSE by 36% (from 0.544 to 0.739), indicating that graded exposure modeling captures contamination magnitude more precisely than a binary flag. The effect on θ-RMSE is moderate (+11%), suggesting that the binary approximation still captures the dominant contamination signal but loses precision on partially exposed items.

Removing judge severity d_j produces minimal degradation (θ-RMSE +1.6%, τ −0.2%). Under the balanced design with $J = 8$ judges, severity effects are averaged out across the fully crossed matrix, reducing the marginal contribution of explicit d_j estimation. We retain d_j in the full model because (i) the MFRM facet specification is theory-driven rather than data-selected, (ii) the d_j profile provides independent diagnostic value for judge calibration (e.g., Claude Opus $d_j = +0.60$ vs. Mistral -0.74 in the real-data evaluation), and (iii) unbalanced or sparse designs would amplify judge severity effects.

D Posterior Predictive Check

Figure 3 shows the posterior predictive distribution from 1,000 posterior draws over 5,661 observations. The predicted and observed response distributions are well-aligned (predicted mean win rate 0.593 vs. observed 0.599), with no systematic discrepancy across item difficulty or model ability levels.

[FIGURE PLACEHOLDER — FIG_PPC.PDF PENDING SYNC FROM REMOTE SERVER]

Figure 3: Posterior predictive check. Distribution of predicted (gray) and observed (black) response rates across all model–item–judge triplets. Mean predicted = 0.593, mean observed = 0.599.

E OVB Validation Across Confounding Levels

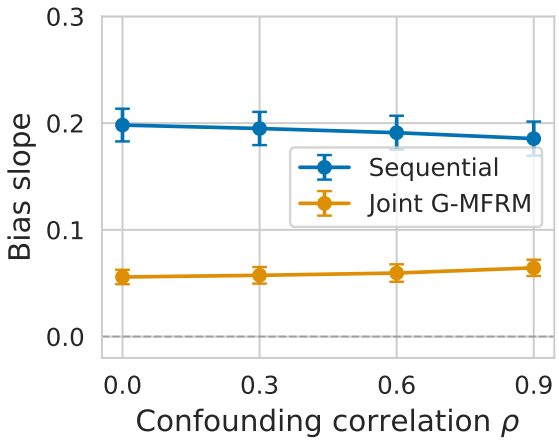


Figure 4: Sequential estimator bias remains approximately constant across confounding levels (ρ), confirming omission-driven rather than confounding-strength-dependent bias. Joint G-MFRM estimation reduces bias by approximately 70% at all ρ levels. Synthetic data: $N = 50$, $I = 100$, $J = 8$, 50 seeds.

F Contamination Susceptibility Recovery

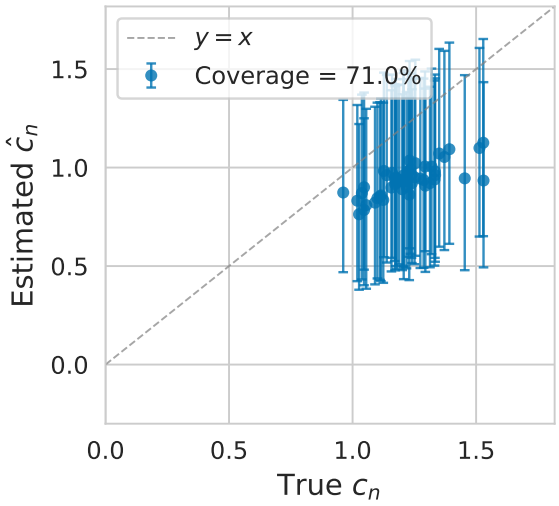


Figure 5: Contamination susceptibility recovery (c_n: true vs. estimated) from 50 Monte Carlo seeds. Empirical CI coverage is 71% (below nominal 95%), indicating moderate posterior overconfidence (see Limitations).

G Error Propagation Figure

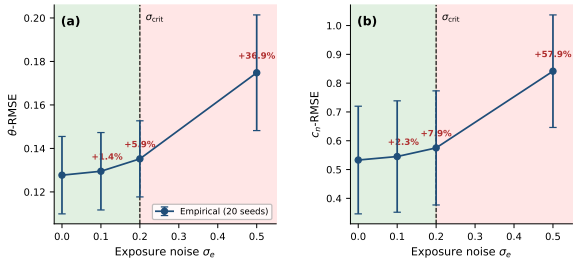


Figure 6: Error propagation from Stage 2 exposure noise to Stage 3 parameter estimates. (a) Ability RMSE (θ) and (b) contamination RMSE (c_n) as a function of exposure noise σ_e. The dashed vertical line marks $\sigma_{\text{crit}} \approx 0.2$, below which degradation remains mild (θ-RMSE $+5.9\%$, c_n-RMSE $+7.9\%$). Error bars show ± 1 std over 20 seeds. See Table 2 for numerical values.